

PunyInform v6.7 quick reference

More information on last page.

Library variables

action

The current action, e.g. `##PutOn`.

action_to_be

The action currently being considered, during parsing.

actor

The creature that is currently being ordered to do something – usually the player.

buffer

The array where player input is kept.

caps_mode

In a `short_name` routine: set to true if object is printed with printing rule (The).

clr_bg

z5+ only: Background colour.

clr_fg

z5+ only: Foreground colour.

clr_fginput PUNY++

z5+ only: Player input colour.

clr_fgstatus

z5+ only: Status line colour.

clr_on

z5+ only: Set to true to enable colours.

consult_from

The first word number of a topic in player input.

consult_words

The number of words of a topic in player input.

deadflag

0 = game is on, 1 = player is dead, 2 = player has won, 3+ = game is over - `DeathMessage()` prints why.

inp1

Usually equals `noun`, but is 1 if `noun` is a number.

inp2

Usually equals `second`, but is 1 if `second` is a number.

input_action PUNY++

The original action, derived from player input.

input_direction PUNY++

The original value of `selected_direction`, derived from player input.

input_noun PUNY++

The original value of `noun`, derived from player input.

input_second PUNY++

The original value of `second`, derived from player input.

inventory_stage

See `invent` property.

inventory_style

0 = tall, 1 = wide. Can be set by game. If `OPTIONAL_FLEXIBLE_INVENTORY` has been defined, player can change it ("inventory tall/wide").

keep_silent

Set to true to make most group 2 actions silent.

location

The room where the player is, or `thedark` if it's dark.

lookmode

1 = normal, 2 = long, 3 = short. Long mode shows the room description on every visit, short never. 1 is default. Many games set it to 2 in `Initialise`.

meta

Is true if action is a meta action, like `##Save`.

no_implicit_actions

Set this to true to disable all implicit actions.

normal_directions_enabled PUNY++

Is normally true. Set it to false to stop the library from recognizing the normal directions (N, S, UP, IN etc).

noun

The primary object of the current action, if any.

num_words

The number of words in player input.

parse

The array to keep track of the words in player input.

parser_action

In a `parse_name` routine, set this to `##PluralFound` if a match was made and it's in plural form.

player

Points to the object that represents the player.

real_location

The room where the player is, even if it's dark.

receive_action

The action (`##Insert` or `##PutOn`) that caused the fake action `##Receive`.

run_after_routines_arg_1 PUNY++

Argument for message (See `run_after_routines_msg`).

run_after_routines_msg PUNY++

Action routine sets to a message# or 1, to run after-routines. 1 means "run after routines, no message". Action routine can also set `run_after_routines_arg_1`.

scope_modified PUNY++

If `OPTIONAL_MANUAL_SCOPE` has been defined, set this to true whenever (possibly) changing what's in scope.

scope_reason ~PUNY

Used in `InScope` routine (but not in `add_to_scope`).

scope_stage

Used in scope token routines referred to in the grammar.

score

The current score.

second

The secondary object of the current action, if any.

selected_direction PUNY++

The direction matched in input, if any, e.g. `n_to`.

ship_directions_enabled PUNY++

If `OPTIONAL_SHIP_DIRECTIONS` is defined, set this to true/false to turn ship directions, e.g. 'aft', on/off.

task_done

A byte array to note which tasks have been completed. Achieved updates this. Requires `TASKS_PROVIDED`.

task_scores

A byte array with scores for tasks. Requires `TASKS_PROVIDED`.

the_time

The current time, in minutes past midnight. Only used for games that show time on the statusline.

turns

The game's turn counter.

update_moved PUNY++

Set this to `true` when using `move [object]` to `player`; (sets moved attribute & updates score).

verb_word

The verb in the current input, e.g. 'take'.

wn

Word number in player input where `NextWord()` will read, from 1 to `(num_words - 1)`.

Library constants

The Inform language defines `true` (1), `false` (0) and `nothing` (0). The library also defines `NULL` (-1), which is used for an action, property or pronoun which currently doesn't have a value. `DIRECTION_COUNT` holds the number of directions recognized: Normally 8, but 12 if `OPTIONAL_FULL_DIRECTIONS` is defined.

`PUNYINFORM_MAJOR_VERSION`,
`PUNYINFORM_MINOR_VERSION`,
`PUNYINFORM_PATCH_VERSION` and
`PUNYINFORM_VERSION_SUFFIX` hold the library version.

There are constants for colours: `CLR_BLACK`,
`CLR_RED`, `CLR_GREEN`, `CLR_YELLOW`, `CLR_BLUE`,
`CLR_MAGENTA`, `CLR_CYAN`, `CLR_WHITE`,
`CLR_DEFAULT`, `CLR_CURRENT`,
`CLR_OZMOO_ORANGE`, `CLR_OZMOO_BROWN`,
`CLR_OZMOO_LIGHT_RED`, `CLR_OZMOO_DARK_GREY`,
`CLR_OZMOO_MEDIUM_GREY`,
`CLR_OZMOO_LIGHT_GREEN`,
`CLR_OZMOO_LIGHT_BLUE`,
`CLR_OZMOO_LIGHT_GREY`.

The global `scope_reason` should always have one of

the values `PARSING_REASON`, `EACH_TURN_REASON`,
`REACT_BEFORE_REASON`, `REACT_AFTER_REASON`,
`LOOPOVERSCOPE_REASON`, `TESTSCOPE_REASON`.

User-defined constants

AMUSING_PROVIDED

Offer the player to see amusing things to try after they complete the game. See `Amusing` entry point routine.

CUSTOM_PLAYER_OBJECT = object PUNY++

The player object to use instead of `selfobj`.

DEATH_MENTION_UNDO ~PUNY

Print 'undo' option even when player wins.

DEBUG

Use debug mode, with debug verbs and more information when something goes wrong.

DEFAULT_CAPACITY = number PUNY++

The number of objects that can be held by a container or supporter that doesn't provide a value for `capacity`.

Headline = "^text^"

Information about the game - genre, author, credits etc. Start with "^", end with "^", separate sections with "^".

INITIAL_LOCATION_VALUE = object PUNY++

The location in which the player starts. A game must either define this, or set `location` in `Initialise`.

MAX_CARRIED = number

Set the `capacity` of the player object (default 32).

MAX_FLOATING_OBJECTS = number PUNY++

The maximum number of objects that can have the `found_in` property (default 32).

MAX_SCOPE = number PUNY++

How many objects can be in scope at once (default 50).

MAX_SCORE = number

The maximum score the player can get.

MAX_TIMERS = number

The maximum number of objects which can have an active timer or daemon at the same time (default 32).

NO_PLACES

Don't define the verbs 'objects' and 'places'.

NO_SCORE = number PUNY++

Don't include a scoring mechanism. In z3 games, show the value of this constant as the score on the statusline.

NUMBER_TASKS = number

The number of scoring tasks (default 1). See `TASKS_PROVIDED`.

OBJECT_SCORE = number

The score for objects with `scored` attribute the player picks up (default 4). Requires `OPTIONAL_SCORED`.

OPTIONAL_ALLOW_WRITTEN_NUMBERS PUNY++

Allow the player to use 'one' .. 'twenty' as numbers.

OPTIONAL_EXTENDED_METAVERBS PUNY++

Enable a set of nice-to-have metaverbs. See "Group 1 actions" in this document.

OPTIONAL_EXTENDED_VERBSET PUNY++

Enable a set of nice-to-have verbs. See "Group 2 actions" and "Group 3 actions" in this document.

OPTIONAL_FULL_DIRECTIONS PUNY++

Enable directions NE, NW, SE, SW.

OPTIONAL_FLEXIBLE_INVENTORY PUNY++

Allow player to change inventory style ("i tall/wide")

OPTIONAL_FULL_SCORE PUNY++

Enable the 'fullscore' verb, and optionally support for tasks (See `TASKS_PROVIDED`).

OPTIONAL_GUESS_KEY PUNY++

Make the parser pick the right key for lock/unlock, if `OPTIONAL_GUESS_MISSING_NOUN` is defined.

OPTIONAL_GUESS_MISSING_NOUN PUNY++

Make the parser fill in missing parts in player input.

OPTIONAL_LANGUAGE_NUMBER PUNY++

Define the `LanguageNumber(n)` routine.

OPTIONAL_LIST_TOGETHER PUNY++

Support the `list_together` property.

OPTIONAL_MANUAL_REACTIVE PUNY++

The author takes responsibility for setting the `reactive` attribute on the right objects.

OPTIONAL_MANUAL_SCOPE **PUNY++**

The author takes responsibility for setting `scope_modified = true` when doing something in code which may affect what's in scope.

OPTIONAL_MANUAL_SCOPE_BOOST **PUNY++**

Avoid running `react_before`, `react_after` and `each_turn` steps whenever possible, for performance.

OPTIONAL_NO_ADD_TO_SCOPE **PUNY++**

Remove support for `add_to_scope` property.

OPTIONAL_NO_DARKNESS **PUNY++**

Rooms are always lit. Don't define the `light` attribute.

OPTIONAL_NON_FLASHING_STATUSLINE **PUNY++**

Use a faster but less compatible statusline (z5+ only).

OPTIONAL_ORDERED_TIMERS **PUNY++**

Timers/daemons are executed in order of their value for property `timer_order`.

OPTIONAL_PRINT_SCENERY_CONTENTS **PUNY++**

Make 'look' describe what's in/on containers/supporters which have the `scenery` attribute.

OPTIONAL_PROVIDE_UNDO **PUNY++**

Enable the 'undo' verb (z5 & z8 only!)

OPTIONAL_REACTIVE_PARSE_NAME **PUNY++**

Use the `reactive` attribute for all objects which have the `parse_name` property.

OPTIONAL_SCORED **PUNY++**

Enable support for the `scored` attribute.

OPTIONAL_SIMPLE_DOORS **PUNY++**

Lets you define property `door_dir` as an array and skip property `door_to` for simple two-way doors.

OPTIONAL_SHIP DIRECTIONS **PUNY++**

Enable support for 'fore', 'aft' etc. Also see variable `ship_directions_enabled`.

OPTIONAL_SL_NO_MOVES **PUNY++**

Don't show moves on statusline (z5 & z8 only!).

OPTIONAL_SL_NO_SCORE **PUNY++**

Don't show score on statusline (z5 & z8 only!).

ROOM_SCORE = number

The score for each room with the `scored` attribute the player visits (default 5). Requires `OPTIONAL_SCORED`.

RUNTIME_ERRORS = number **PUNY++**

What to do when a runtime error occurs: 2 = Print error message, 1 = Print error code, 0 = Like 1 but also reduce checks for errors (default: 2 if `DEBUG` is defined, 1 if not)

SACK_OBJECT = object

The object the player will automatically try to put belongings into when the player's `capacity` has been reached.

STATUSLINE_SCORE **PUNY++**

Show score and moves, not time on the statusline.

STATUSLINE_TIME **PUNY++**

Show time, not score on the statusline.

Story = "text"

Mandatory: The name of the game.

TASKS_PROVIDED

Use tasks for scoring. Also define `NUMBER_TASKS`, byte array `task_scores` and entry point routine `PrintTaskName`.

Library routines

Achieved(number)

Complete task `number`. Requires `TASKS_PROVIDED`.

Banner()

Print the game name, serial# etc. To skip this at game start, return 2 from `Initialise` and call `Banner` later.

ChangeFgColour(colour) **PUNY++**

Change the foreground colour, if game is in z5+ format and colour has been enabled.

ChooseObjectsFinal_Discard(number) **PUNY++**

Call from entry point routine `ChooseObjectsFinal` to remove option number (0 or higher).

ChooseObjectsFinal_Pick(number) **PUNY++**

Call from entry point routine `ChooseObjectsFinal` to pick option number (0 or higher).

ClearScreen(window) **~PUNY**

z5+ only. Call with argument `WIN_MAIN`, `WIN_STATUS` or `WIN_ALL` to set colours (if `clr_on` is true) and clear main and/or status window.

CommonAncestor(object1, object2)

Return the nearest object that contains both `object1` and `object2` on some level, or false.

DoorDir(object) **PUNY++**

Returns the direction in which the door object lies.

DoorTo(object) **PUNY++**

Returns the room which the door object leads to, or 1 if `object.door_to` is a routine and it returned 1, meaning it printed a message why the player can't enter.

DrawStatusLine(newlines) **~PUNY**

z5+ only. Print the statusline. If `newlines` is true, it must instead print as many newlines as the statusline height

FastSpaces(number) **PUNY++**

Print `number` spaces in an efficient manner.

ImplicitDisrobeIfWorn(object) **PUNY++**

Take off object if worn. Return false for success.

ImplicitGrabIfNotHeld(object) **PUNY++**

Take object if not held. Return false for success.

IndirectlyContains(object1, object2)

Return true if `object1` holds `object2`, on some level.

IsARoutine(value) **PUNY++**

Return true if `value` can be a routine address.

IsAString(value) **PUNY++**

Return true if `value` can be a string address.

LanguageNumber(number)

Print `number` using words. Requires `OPTIONAL_LANGUAGE_NUMBER`.

LoopOverScope(routine, actor)

Call routine once for each object in scope for `actor` (default: `player`), passing the object as a parameter.

MoveFloatingObjects()

Check which objects with `found_in` should be present.

NextWord()

Read word at position `wn` in player input and increment `wn`. Return the word, or false if empty or unrecognized.

NextWordStopped()

Like `NextWord()` except it returns `-1` if `wn` points beyond the end of player input.

NumberWord(numword)

If `numword` is a word, e.g. 'six', representing a number 1-20, return the number. If not, return false. Requires `OPTIONAL_ALLOW_WRITTEN_NUMBERS`.

NumberWords()

Return the number of words in player input.

ObjectCapacity(object) PUNY++

Return the object's value for the `capacity` property, defaulting to `DEFAULT_CAPACITY` or 100 if not defined.

ObjectIsUntouchable(object, flag)

Return true if player can't touch `object`. If so, and `flag` is false or unspecified, also print a message.

OzmooColoursAvailable() PUNY++

Returns true if the game is being played on Z-code interpreter Ozmoo, with the 8 extra colours available.

ParseToken(type, data)

Use in general parsing routines. See DM4 index for examples.

PlaceInScope(object)

Use in user-supplied scope routines to put `object` in scope. Ignores `add_to_scope` property and children.

PlayerTo(object, flag)

Move player to `object`, which must be a room or an enterable object. If `flag` is 0 or omitted, print a long room description. If `flag` is 1, keep quiet. If `flag` is 2, print a room description based on the `lookmode` value.

PrintContents(text, object, style) PUNY++

Recursively list contents of `object`. Hide items that have concealed or scenery, unless `action` is `##Inv`. Print or run `text` (unless 0) before first item. If `style` has `WORKFLAG_BIT` set, only print objects which have the `workflag` attribute. If it has `ISARE_BIT` set, print "is" or "are" before list. If it has `NEWLINE_BIT` set, print each object on a new line. Return true if any objects were

listed. Call with `text==1` to not print anything but return 0 if `object` contains no printable objects, 1 if contents can be prefixed with "is", 2 for "are".

PrintContentsFromR(text, object) PUNY++

Like `PrintContents`, but list `object` and its siblings + contents, retaining current style and indentation.

PrintMsg(msg, arg1, arg2) PUNY++

Print library message `msg`. Some messages need an argument or two, use `arg1` and `arg2` for this.

PrintOrRun(object, property, flag)

If `object.property` holds a routine or list of routines, run them using `RunRoutines(object, property)`. If it holds a string, print it and then a newline. If `flag` is true, skip the newline.

PronounNotice(object)

Make a pronoun ('it', 'her' etc) refer to `object`.

RunRoutines(object, property, switch)

If `object.property` holds a routine or list of routines, run each routine until one of them returns a non-zero value. Return the return value of the last routine run. If `switch` has a non-zero value, the routines can have switch-clauses to match this value, otherwise they can have switch-clauses to match `action`.

ScopeWithin(object)

Use in user-supplied scope routines. Put all items which are in `object` into scope, also recursively searching supporters, transparent objects and open containers, and check the `add_to_scope` property of all objects added.

SetTime(number, step)

Sets the time to `number` minutes after midnight. If `step` is a positive number, one turn takes `step` minutes. Otherwise, there are `-step` turns to a minute.

TestScope(object, actor)

Return true if `object` is in scope for `actor` (defaults to player).

TryNumber(wordnum)

Try to parse word `wordnum` in player input as a number. Return the number (0-10000), or -1000 for invalid number. If `OPTIONAL_ALLOW_WRITTEN_NUMBERS` is defined, also parse number words ('one' .. 'twenty').

WordAddress(wordnum)

Return the address in memory where the characters for word `wordnum` in player input are stored.

WordLength(wordnum)

Return the length of word `wordnum` in player input.

WordValue(wordnum)

Return the dictionary word that word `wordnum` in player input matches, or 0 if no match was found.

YesOrNo()

Wait for the player to type something. Return true if they typed yes, false if they typed no, or ask again.

Printing rules

A printing rule is used to print something based on one argument, typically an object. Example of use:
`print "The pump is ", (OnOff) Pump, ".^";`

CObjIs PUNY++

Prints "The (object) ", and "is" or "are" (see `IsOrAre`).

CTheyreorIts PUNY++

Prints "It's" or "They're" or "He's" etc.

CTheyreorThats

Prints "That's" or "They're" or "He's" etc.

IsOrAre

Prints "is" or "are", based on `pluralname` and if the object is the player object.

ItorThem

Prints "it", "them", "her" etc.

ObjIs PUNY++

Prints "the (object) ", and "is" or "are" (see `IsOrAre`).

OnOff PUNY++

Prints "on" or "off", based on `on`.

SingularS PUNY++

Prints "s" if the object doesn't have `pluralname`.

ThatorThose

Prints "that" or "those" based on `pluralname`.

Entry point routines

These routines, if defined by the game author, are run under the circumstances stated for each routine.

`AfterLife()`

When player has died. Can be used to revive player.

`AfterPrompt()`

After the input prompt has been printed.

`Amusing()`

When the player has won. Use it to print fun facts about the game. Requires `AMUSING_PROVIDED`.

`BeforeParsing()`

After player input, before parsing starts.

`ChooseObjects(obj, code)`

If `code` is 2, return a score 0-9 for how good a fit `obj` is for action `action_to_be`. Code 0 and 1 means player has typed ALL and parser means to exclude (0) or include (1) the object. Return 0 to don't interfere, 1 to force inclusion, 2 to force exclusion.

`ChooseObjectsFinal(array, length) PUNY++`

Called just before having to ask the player which object they mean. `array` holds the address of a word array holding object numbers, `length` holds the number of objects. Use `ChooseObjectsFinal_Pick` and/or `ChooseObjectsFinal_Discard` to either pick an object or discard options. You may rearrange the array.

`DarkToDark()`

When player moves from one dark location to another.

`DeathMessage()`

When game ends and `deadflag` is set to 3 or higher. Should print a few words to say why the game ended.

`DebugParseNameObject(object) PUNY++`

When the parser checks for matching objects for a debug verb like 'purloin'. Return true if `object` has a `parse_name` routine.

`DisallowTakeAnimate() PUNY++`

When the player tries to take `noun`, which has `animate`. Return false to allow this, or true to disallow it.

`GamePostRoutine()`

When after routines have been run.

`GamePreRoutine()`

Before before routines have been run.

`InScope(actor)`

When working out the scope for the actor. Call `ScopeWithin` and `PlaceInScope` to add objects to scope. Return true if no other objects should be in scope.

`Initialise ~PUNY`

Mandatory: A (possibly empty) routine which is called when the game starts. May print an introduction. May return 2 to skip the game banner. Must set `location`, unless `INITIAL_LOCATION_VALUE` is defined.

`IsARoom(obj, verdict) PUNY++`

Player has typed PLACES, GOTO or ROOMS. `Verdict` = 1 or 0 says if library thinks `obj` is a room or not. Return 0 to accept, 1 to say it's a room, 2 to say it's not.

`LibraryMessages(number, arg1, arg2) PUNY++`

When a library message is about to be printed. Use it to print your own complex library messages.

`LookRoutine`

After the room and everything in it has been described.

`NewRoom`

When the player has entered a new room, before the room is described.

`ParseNoun(object) ~PUNY`

When checking if input matches `object`, before `parse_name` and `name` properties are checked. Can advance `wn` and return -1 to consume words (`parse_name + name` will also be checked), just return -1 to not interfere, or return how many words matched.

`ParseNumber(buffer, length)`

When the parser needs to check if the input word that starts at `buffer` and is `length` bytes long, is a number. Return the number, or false if no number was found.

`PrintRank()`

When the scoring message is printed. Prints the final part, typically giving the player a rank based on `score`.

`PrintTaskName(n)`

When listing a completed task. Print name of task `n`.

`PrintVerb(verb)`

When the parser needs to print a verb. Typically needed for long verbs. Return true if the routine printed the verb.

`TimePasses()`

After a game turn has ended, in which `turns` increased.

`UnknownVerb(word)`

When the parser doesn't recognize the verb. Return a dictionary word to use as the verb instead, or false.

Library objects

`Directions PUNY++`

Represents all directions. The `selected_direction` variable holds the matched direction (e.g. `n_to`) or 0.

`selfobj`

The default player object. It's better to use the `player` variable which normally refers to this.

`thedark`

Fake room. When `real_location` is dark, `location` points here. The player is never moved to `thedark`.

Group 1 actions

Group 1 actions are metaverbs that control gameplay, and debug verbs. They don't run before or after routines.

Again, FullScore, LookModeLong, LookModeNormal, LookModeShort, NotifyOff, NotifyOn, Oops, OopsCorrection, Restart, Restore, Save, Score, Version, Quit

OPTIONAL_EXTENDED_METAVERBS adds:

CommandsOff, CommandsOn, CommandsRead, Objects, Places, ScriptOff, ScriptOn, Verify

DEBUG adds: ActionsOff, ActionsOn, Debug, Forest, GoNear, Goto, Pronouns, Purloin, RandomSeed, Rooms, RoutinesOff, RoutinesOn, Scope, TimersOff, TimersOn, Tree

Group 2 actions

These actions change something in the game world or print information about it. They run before and after routines.

Close	"CLOSE (noun)"
Disrobe	"TAKE OFF (noun)"
Drop	"DROP (noun)"
Eat	"EAT (noun)"
Enter	"ENTER (noun)"
Examine	"EXAMINE (noun)"
Exit	"EXIT (noun)"
GetOff	"GET OFF (noun)"
Go	"GO (direction)"
Insert	"INSERT (noun) INTO (second)"
Inv	"INVENTORY"
Lock	"LOCK (noun) with (second)"
Look	"LOOK"
Open	"OPEN (noun)"
PutOn	"PUT (noun) ON (second)"
Remove	"REMOVE (noun) FROM (second)"
Search	"SEARCH (noun)"
SwitchOff	"SWITCH OFF (noun)"

SwitchOn	"SWITCH ON (noun)"
Take	"TAKE (noun)"
Transfer	"TRANSFER (noun) TO (second)"
Unlock	"UNLOCK (noun) WITH (second)"
Wait	"WAIT"
Wear	"WEAR (noun)"

OPTIONAL_EXTENDED_VERBSET adds:

Empty	"EMPTY (noun)"
EmptyT	"EMPTY (noun) INTO (second)"
GoIn	"INSIDE"

Group 3 actions

These actions normally just print a standard message. They run before routines but not after routines.

Answer	"ANSWER (topic) TO (second)"
Ask	"ASK (noun) ABOUT (topic)"
AskFor	"ASK (noun) FOR (second)"
AskTo	"ASK (noun) TO (topic)"
Attack	"ATTACK (noun)"
Climb	"CLIMB (noun)"
Consult	"READ ABOUT (topic) IN (noun)"
Cut	"CUT (noun)"
Dig	"DIG (noun)"
Drink	"DRINK (noun)"
Fill	"FILL (noun)"
Give	"GIVE (noun) to (second)"
Jump	"JUMP"
JumpOver	"JUMP OVER (noun)"
Listen	"LISTEN TO (noun)"
Pull	"PULL (noun)"
Push	"PUSH (noun)"
PushDir	"PUSH (noun) (direction)"
Rub	"RUB (noun)"
Shout	"SHOUT (topic)"
ShoutAt	"SHOUT (topic) AT (second)"
Show	"SHOW (noun) (second)"
Smell	"SMELL (noun)"
Tell	"TELL (noun) ABOUT (topic)"

ThrowAt	"THROW (noun) AT (second)"
Tie	"TIE (noun) TO (second)"
Touch	"TOUCH (noun)"
Turn	"TURN (noun)"

OPTIONAL_EXTENDED_VERBSET adds:

Blow	"BLOW (noun)"
Mild	"DARN"
Burn	"BURN (noun)"
Buy	"BUY (noun)"
Kiss	"KISS (noun)"
No	"NO"
Set	"SET (noun)"
SetTo	"SET (noun) TO (special)"
Strong	"SHIT"
Sing	"SING"
Sleep	"SLEEP"
Sorry	"SORRY"
Squeeze	"SQUEEZE (noun)"
Swim	"SWIM"
Swing	"SWING (noun)"
Taste	"TASTE (noun)"
Think	"THINK"
Transfer	"TRANSFER (noun) TO (second)"
Wake	"WAKE UP"
WakeOther	"WAKE UP (noun)"
Wave	"WAVE"
Yes	"YES"

Fake actions

These actions are not referred to anywhere in the grammar, and they don't have action routines, e.g. the fake action Going has no action routine GoingSub.

Going	Sent to the before routine for the room that the player is about to enter.
LetGo	Sent to the container/supporter from which the player takes something.
NotUnderstood	Sent to creature's orders when player issued an incomprehensible order to it.

Order	Sent to creature's life when player issued an order to it, and <code>orders</code> didn't handle it.
PluralFound	A <code>parse_name</code> routine can set <code>parser_action</code> to this value when a match is found and it's in plural.
Receive	Sent to the object the player tries to place something in/on. <code>receive_action</code> holds the original action.
ThrownAt	Sent by action <code>ThrowAt</code> to the object the player tries to throw something at.

Object attributes (flags)

An attribute is a flag which can be on or off.

[OBJ] means this is used for regular objects.

[ROOM] means this is used for rooms.

`absent` [OBJ]

For object with `found_in`: Removed from game for now.

`animate` [OBJ]

Is a living thing, can be talked to etc.

`clothing` [OBJ]

Can be worn.

`concealed` [OBJ]

Is visible but not easy to spot, like a secret door. Can be interacted with but is not printed in room description.

`container` [OBJ]

Objects can be put in it and removed from it, if it's open.

Can't also have `supporter`. Can have `enterable`.

`door` [OBJ]

Is a portal between rooms. Use properties `door_to`, `door_dir` and, unless it's a one-way door, `found_in`.

`edible` [OBJ]

Can be eaten.

`enterable` [OBJ]

Can be entered. Must have `container` or `supporter`.

`female` [OBJ]

Can be referred to as she/her. Must have `animate`.

`general` [OBJ] [ROOM]

To be used by the game author for whatever they like.

`light` [OBJ] [ROOM]

Provides light. For room and container, lights up what's inside as well. Note: This attribute is not defined if `OPTIONAL_NO_DARKNESS` is defined.

`lockable` [OBJ]

Can be locked and unlocked, using the object specified by `with_key` property.

`locked` [OBJ]

Can't be opened.

`moved` [OBJ]

Is or has been held directly by the player.

`neuter` [OBJ]

Can be referred to as "it" (Mainly used for `animate` objects, as this is default behaviour for non-animates).

`on` [OBJ]

Is currently switched on. See `switchable` attribute.

`open` [OBJ]

For doors and containers: Is currently open.

`openable` [OBJ]

For doors and containers: Can be opened and closed.

`pluralname` [OBJ]

Can be referred to as they/them.

`proper` [OBJ]

Has a name which should never be preceded by an article, like "John".

`reactive` [OBJ] [ROOM] **PUNY++**

The object provides at least one of `add_to_scope`, `each_turn`, `react_before`, `react_after` (+ `parse_name` if `OPTIONAL_REACTIVE_PARSE_NAME` is defined). Note: unless `OPTIONAL_MANUAL_REACTIVE` is defined, the `reactive` attribute is set automatically.

`scenery` [OBJ]

Can't be taken, is not mentioned in room descriptions.

`scored` [OBJ] [ROOM]

For an object: awards `OBJECT_SCORE` points when taken for the first time. For a room: awards `ROOM_SCORE` points when visited for the first time. Note: Only defined if `OPTIONAL_SCORED` is defined and `NO_SCORE` is not.

`static` [OBJ]

Can't be taken.

`supporter` [OBJ]

Is a supporter, meaning things can be placed on top of it. Can't also have `container`. Can have `enterable`.

`switchable` [OBJ]

Can be switched on and off. The `on` attribute tells its current state.

`talkable` [OBJ]

Can be talked to, even though it's not `animate`.

`transparent` [OBJ]

For a `container`: The contents are visible even if the container is closed. For an `animate` object: Held objects are visible to others. For other objects: Objects that are part of this objects (i.e. are *inside* this object) are visible.

`visited` [ROOM]

The player has seen this room.

`workflag` [OBJ] [ROOM]

Temporary internal flag. Can be used by game code too.

`worn` [OBJ]

For object that has `clothing`: Is currently being worn.

Object properties

A property is a 16-bit value or a list of values.

[OBJ] means this is used for regular objects.

[ROOM] means this is used for rooms.

(+) means "additive" - if an object which defines the property inherits from a class which also defines the property, it gets both values.

add_to_scope [OBJ]

A list of objects that should be added to scope when this object is in scope, or a routine which puts objects in scope using `ScopeWithin` and `PlaceInScope`.

after [OBJ] [ROOM] (+)

For an object: Receives every action and fake action for which this is the noun.

For a room: Receives every action which occurs here.

The property value is a routine, which usually has sections like switch-clauses, each listing one or more actions, a colon and the code to run. There can be a `default` clause which runs if nothing else was matched. There can also be code before the first clause, which will run regardless of action. The routine should return `false` to continue, telling the player what has happened, or `true` to stop processing the action and produce no further output.

article [OBJ] ~PUNY

A string or a routine to print the indefinite article for the object name. The default article is "some" for objects that have `pluralname`, nothing for objects that have `proper`, and "a" for all others.

before [OBJ] [ROOM] (+)

Like `after`, but is run before the action happens.

Returning `true` stops the default action from happening at all.

cant_go [ROOM]

A string or a routine to print a message, when the player tries to go in a direction where there's no exit.

capacity [OBJ] ~PUNY

The maximum number of items that can be in this container, on this supporter or held by this actor. To read the capacity of an object, taking the default capacity into

consideration, you must call `ObjectCapacity(object)`.

d_to [ROOM]

Holds a possible exit. The value can be any of:

- * `false` – not an exit
- * a room where the exits leads
- * a door object – the exit leads through this door
- * a string saying why the player can't go there
- * a routine which either returns `false`, a room, a door object, or prints its own message and returns `true`.

daemon [OBJ] [ROOM]

A routine that is executed every turn once it is started. Use `StartDaemon` and `StopDaemon` to start/stop it.

describe [OBJ] (+) ~PUNY

A string or a routine to print a paragraph of text for an object in a room description. If it's a string, the value 1 or 2, or a routine which returns 1 or 2, the object won't be further described. If the value / return value is 2, the library will consider it as if nothing was printed. Not supported for rooms as in DM4. Note: Start and end the description with a newline ("
").

description [OBJ] [ROOM]

For an object: A string or a routine to print the text the player should get when examining the object.
For a room: A string or a routine to print the room description.

door_dir [OBJ] ~PUNY

For a door: A direction (e.g. `n_to`) or a routine returning a direction. This says in which direction the door lies in `location`. If `OPTIONAL_SIMPLE_DOORS` is defined and `found_in` holds a list with two rooms, `door_dir` can be a list of two directions.

door_to [OBJ] ~PUNY

For a door: A room or a routine returning a room. This says where the door leads, when the player is in `location`. If `OPTIONAL_SIMPLE_DOORS` is defined and `found_in` holds a list with two rooms, `door_to` can be omitted.

e_to [ROOM]

An exit property. See `d_to`.

each_turn [OBJ] [ROOM] (+)

A routine which is executed every turn when the object is in scope.

found_in [OBJ]

A list of rooms where the object is present, or a routine which returns `true` if the object is present in `location`. If the object has `absent`, it's not present anywhere.

grammar [OBJ] ~PUNY

For `animate` or `talkable` objects: Called when object is spoken to. Can advance `verb_wordnum`. Return `true` if routine has parsed all input and set up `action`, `noun` and `second`. Return 'verb' and set `parser_one` = 1 to use this verb's grammar instead, or return '-verb' and set `parser_one` = -1 to use this verb's grammar but fall back to the verb in player input if parsing fails. Return `false` to parse as usual.

in_to [ROOM]

An exit property. See `d_to`.

initial [OBJ] [ROOM]

For an object: A string or a routine to describe the object before it's been picked up. Note: For doors/containers and switchable objects, use `when_open` + `when_closed` and `when_on` + `when_off` respectively.
For a room: A string or a routine to print a text when the player enters the room.

inside_description [OBJ]

For an enterable object: A string or a routine that will printed/run when the player is in/on the object.

invent [OBJ]

A routine to print the object in a list (typically in player inventory or a room description). First the routine is called before the object name has been printed, with `inventory_stage` set to 1. Then it's called again when the object name has been printed but no additional information (e.g. "(providing light)"), with `inventory_stage` set to 2.

For both calls, the routine should return `false` to continue or `true` to stop all further output.

life [OBJ] (+)

For `animate` objects: Works like a `before` routine, but receives only person-to-person actions (`Answer`, `Ask`,

Attack, Give, Kiss, Order, Show, Tell, ThrowAt, WakeOther). Can be a string instead of a routine.

list_together [OBJ] ~PUNY

Objects that have the same value are listed together in object listings. Set to a low number to print them adjacent to each other, a string to print them as a group, or (advanced) a routine to handle the printing yourself. Can't handle identical objects. Requires OPTIONAL_LIST_TOGETHER.

n_to [ROOM]

An exit property. See **d_to**.

name [OBJ] [ROOM]

A list of dictionary words. For an object, these are the words that can be used to refer to the object. For a room, these are words which should yield a reply like "You don't need to refer to that.". For an object, but not a room, the **name** property can be overridden by the **parse_name** property.

ne_to [ROOM]

An exit property. See **d_to**. Requires OPTIONAL_FULL_DIRECTIONS to work.

nw_to [ROOM]

An exit property. See **d_to**. Requires OPTIONAL_FULL_DIRECTIONS to work.

orders [OBJ]

For **animate** or **talkable** objects: A routine to carry out the player's orders or decline to do so. The routine should either return **false**, or print a message and return **true** to stop further processing. The player object's **orders** routine is called first, and then the addressed object's **orders** routine is called.

out_to [ROOM]

An exit property. See **d_to**.

parse_name [OBJ]

A routine to parse player input and decide if it matches this object. The routine calls **NextWord()** and/or **NextWordStopped()** to read words and returns the number of words that match, 0 for no match or -1 to say it chooses not to decide (i.e. the **name** property will be consulted, if provided).

react_after [OBJ]

Like **after**, but receives all actions taking place when this object is in scope.

react_before [OBJ]

Like **before**, but receives all actions taking place when this object is in scope.

s_to [ROOM]

An exit property. See **d_to**.

se_to [ROOM]

An exit property. See **d_to**. Requires OPTIONAL_FULL_DIRECTIONS to work.

short_name [OBJ]

A string or routine to print the short name of the object, overriding the name provided in the object's name string. The routine should return **true** to signal that it has printed the name, or **false** to say that the library should still print the object's name string. Sometimes it's useful to print a prefix (e.g. an adjective) and return **false**.

sw_to [ROOM]

An exit property. See **d_to**. Requires OPTIONAL_FULL_DIRECTIONS to work.

time_left [OBJ] [ROOM]

For an object which has a **time_out** property: After **StartTimer(object, turns)** has been called, **time_left** holds the number of turns left before **time_out** will be called.

For other objects/rooms: Use it as a general variable.

time_out [OBJ] [ROOM] (+)

A routine to be called when a timer times out. Start the countdown with **StartTimer(object, turns)**.

timer_order [OBJ] [ROOM] PUNY++

A number that determines when this object's **timer/daemon** executes relative to other **timers/daemons**. The lower the earlier. Objects that don't provide it have the value 100. Requires OPTIONAL_ORDERED_TIMERS.

u_to [ROOM]

An exit property. See **d_to**.

w_to [ROOM]

An exit property. See **d_to**.

when_closed [OBJ]

For doors and containers: A string or routine to describe the object when it's closed.

when_off [OBJ]

For a switchable object: A string or routine to describe the object when it's off.

when_on [OBJ]

Like **when_off**, but for when it's on. Not used if the object has moved.

when_open [OBJ]

Like **when_closed**, but for when it's open. Not used if the object has moved.

with_key [OBJ] ~PUNY

For **lockable** objects: The object which works as a key, or a routine which returns **true** if the object held in **second** works as a key.

About this document

This is meant to be printed out and serve as a quick index to all the functionality that the PunyInform library provides. A similar document for the Inform 6 language is **Inform in Four Minutes**, available at <http://www.firthworks.com/roger/>

This is not meant to be a document from which to learn PunyInform, a replacement for The Inform Designer's Manual, Fourth Edition (DM4) or the PunyInform manual and tutorials. To keep it short, this document leaves out finer details. Always consult DM4 and/or the PunyInform manual to get the full picture (See Legend below).

Created and maintained by Fredrik Ramsberg. Improvement suggestions by Garry Francis, Johan Berntsson and Nick Moffitt.

Based on InfoLib at Your Fingertips by Roger Firth.

Legend

Items marked **PUNY++** aren't described in DM4. Items marked **~PUNY** don't work exactly as described in DM4. See PunyInform manual for details on these items.